

Climate Resilience Multi-Phased Programmatic Approach (CResMPA)

TERMS OF REFERENCE

Consultant Service for designing and developing a Decision Support System (DSS) to enhance Emergency Preparedness and Response for Disaster Risk Reduction.

Background

The Government of Sri Lanka (GoSL) with support from the World Bank (WB) is currently in the process of implementing an investment project titled, the "Climate Resilience Multiphase Program (CResMPA)." This program is executed by the Ministry of Irrigation with the implementation support of the Ministry of Defence.

The main objective of component 1 of the project is to strengthen the capacity of the GoSL to deliver effective forecasts and warnings of floods and landslides for early action. The project involves four agencies: the Department of Meteorology (DoM), the Irrigation Department (ID), the National Building Research Organization (NBRO), and the Disaster Management Center (DMC). These terms of reference (TOR) focus on establishing DSS. This TOR is simultaneously interconnected with upgrading the Disaster Management Centre (DMC) facilities to improve the community response.

A key component of the project is the design and implementation of a Decision Support System (**DSS**). The **DSS** will provide a centralized platform to integrate hazard, risk, and socioeconomic data, enabling timely and informed decision-making. This system aims to improve situational awareness, optimize resource allocation, and enhance coordination among agencies to adapt and mitigate disaster impacts effectively.

The DSS will feature advanced technologies, including real-time geospatial analytics, predictive modelling, and AI-driven insights. It will serve as a critical tool for anticipating climate risks, supporting emergency responses, and fostering long-term resilience in vulnerable communities.

This initiative represents a significant step towards aligning Sri Lanka's disaster management capabilities with international best practices, ensuring a proactive and data-driven approach to disaster preparedness and response. The Decision Support System provides a contextual analysis of a variety of hazards, risks and vulnerabilities, and socioeconomic information to support informed decision-making.

One of the key challenges in disaster response is to ensure that timely information reaches related agencies and first responders before they make critical decisions. The first few hours following a disaster are when key decisions are made about decision assessment and resource allocation. Without timely warnings and information, rescuers lose critical time for important tasks such as evacuation, locating clusters of probable survivors, and, more generally, the optimization of response and recovery options.

A smart geospatial platform performs real-time collection, management, analysis, distribution, and visualization of information for enhanced situational awareness. A smart platform can visually display the disaster and its location accurately in real-time, hence users can more rapidly understand the disaster's distribution and predicted trajectory. This real-time stream of critical information populates the optimisation/simulation engine which increases the cognitive abilities of decision-makers when faced with disasters of large magnitude and uncertainty.

This will serve as the digital interface for collaboration among all primary agencies and other stakeholders at the national and local levels; enabling the development of impact scenarios to anticipate and respond early to emerging risks associated with climate change and the evolving vulnerabilities of society and the economy; and provide an education and training environment to improve anticipatory action.

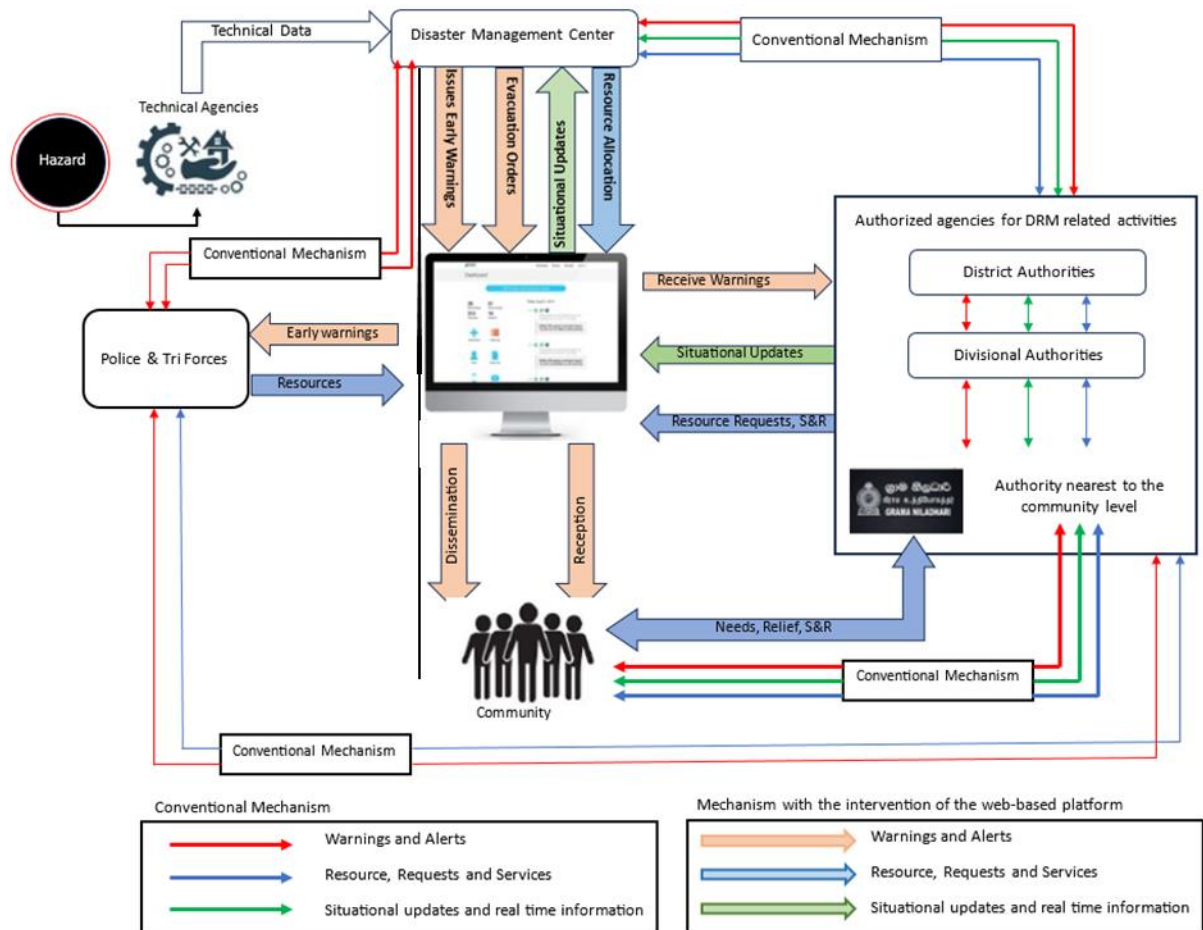


Figure 1 Proposed DSS Flow Chart

The Overview of the Project:

This assignment must align with the World Bank's goals of fostering climate resilience and disaster risk reduction as outlined in the Sendai Framework for Disaster Risk Reduction 2015–2030 and the Paris Agreement. By leveraging advanced technologies such as predictive analytics, AI-driven models, and geospatial visualization, the Decision Support System (DSS) will strengthen institutional capacity and community preparedness. The system aims to minimize disaster impacts, safeguard vulnerable populations, and promote sustainable development by integrating the global best practices for disaster resilience into Sri Lanka's national framework. This approach ensures the project contributes to the World Bank's overarching mission of reducing poverty and building shared prosperity in a sustainable manner.

Objectives of the Project:

- a. Monitor real-time parameters against the standard environmental parameters in Sri Lanka,
- b. Integrate information/ data flows related to predicting disasters for issuing early warnings.
- c. Perform as a high-level operational ICT platform for receiving and sharing/ dissemination of information, data, and functions of different stakeholders
- d. Capture the pre-deployment project activity that attracts Subject Matter Expert (SME) knowledge that must be sorted during multiple project phases.
- e. Integrate real-time and historical environmental data for predictive analytics and risk management.
- f. A real-time data dashboard for risk/data visualization and quick decision-making.
- g. Incorporate predictive models for anticipatory actions and decision support.
- h. Implement a software to modular, and plug-able architecture for scalability.
- i. Strengthen resource management through enterprise resource Planning (ERP) systems for Operational, logistics, Planning, financial and Asset tracking.
- j. Capable of supporting seamless communication and integration with Unmanned Aerial Vehicle -Software Define Networks (UAV-SDN), IOT sensors, and other field data collection tools.
- k. Enhance situational awareness through GIS visualizations.
- l. Community-driven and User-friendly software.
- m. All stakeholders/public visualization and feedback mechanism.

Role of the DSS during a disaster

The DSS system will acquire and process the data received from the agencies, and process according to a predefined set of rules to assist and automate/ coordinate a range of corrective measures taken by the DMC including notification, alerting functions, escalations, and coordination.

Key Function of the DSS

A. Risk Identification :

- a. DSS will perform the anticipatory action by predicting risks based on the agency data (Both historical data, and real-time data)

B. Issue Reporting and Status monitoring :

- b. DSS will aggregate the relevant stakeholders and service providers who will be part of the situation management team formed to manage a current situation, and coordinate communication via a streamlined approach.

- c. DSS will track the progress, measure actions against pre-agreed SLAs (Service Level Agreements), escalate to ensure the actions taken are reported to the relevant parties, and produce required reporting to assist with the closure of the situation.

B. Continuous Improvements:

- a. DSS will provide Analytical reporting on case performance, and how SLAs with stakeholders, can be further tuned to optimize the internal processes, and the DSS system's pre-defined decision rules.
- b. DSS will also integrate several internal processes executed by the Disaster Management Center to ensure the seamless transition of information between divisions to improve communication, effective management of information, and after-response times.

C. Modular Architecture:

- a. The DSS will adopt a world-class architecture. *It is required to follow Digital Public standards.*
- b. **Containerized System:** Containerized software system using such that the system is an Operating System (OS) and on-premises hosting system.
- c. **Front End Content Management:** Highly secure and modular framework.
- d. **Backend ERP:** Software for resource planning, fiscal management, and operational workflows.
- e. **GIS Mapping:** Data and information visualization facilities should be in cooperation.
- f. **Streaming Integration:** Real-time data collection and integration facilities.
- g. **Visualization:** Advanced dashboards for real-time reporting and situational awareness using Data Visualization Frameworks (DVF)

D. Predictive Analytics status

- a. Predictive Risk Analysis and Early Warning: Machine learning models for disaster risk detection.
- b. **Resource Optimization:** Automating the allocation of resources based on data-driven insights.
- c. Automated reporting, data summarization, and communication with field responders.
- d. **Simulation and Forecasting:** Simulations help estimate the impact of potential disasters under various conditions.

E. Enhanced Communication and Security

- a. **UAV-SDN Integration:** Integration of UAVs and Software-Defined Networking (SDN) for field connectivity.
- b. **Cybersecurity Measures:** Existing and next-generation security system and monitoring tools to protect system integrity.
- c. **Resilience:** Ensure communication and operational continuity.

Environmental Status Data from the Agencies, and Staff

The information that will be integrated will primarily be received from technical agencies such as the Department of Meteorology (DoM), the Irrigation Department (ID), the National Building Research Organization (NBRO) etc. Around 50 agencies will be involved in the system based on the National Emergency Operation Plan (NEOP) and NEOP is freely available to visit by the consultant at www.dmc.gov.lk. The consultant shall assess the mission requirement of each agency and accommodate that information and ongoing activities for direct numerical control (DNC) management. Vulnerability assessment and data sets stated below are available at DMC, and consultants can visit those documents. Data sets available at DMC;

1. Disinventar- historical data
2. Hazard profile
3. Early warning messages./ situation reports/ warning signs

The DMC's staff based on their delegation levels will also be able to provide data manually into the DSS system, which will then be used to update the status of an ongoing situation or for usual ongoing risk management purposes. DSS will be operated as a standalone system which is operated and maintained by the DMC interconnected with other technical agencies and relevant stakeholders.

The DSS will be configured periodically to adjust the information processing rules, and actions as the agencies and the environmental parameters change.

DSS will also communicate with the agencies concerning the information received periodically for ongoing risk assessments and will assess the information received during a situation; and will provide analytical predictions and recommendations on continuous improvements.

Response teams and 3rd party service providers.

DSS will coordinate efforts using effective communication, based on predefined rules, and on-demand requests from Situation Assessment officers. The Communication pathways, workflow sequences, and escalations will also be part of the SLAs recorded in the DSS system's rules engine, which will ensure appropriate service providers, and their contingency resources are contacted.

Feedback from ground staff, for real-time monitoring, and action.

DSS will receive a range of information from its regional/national/local operations teams, and offices. The information types and the actions taken by the DSS will ensure the system's flexibility to adapt and respond fast and provide up-to-date information to the DSS system's automated decision system, and to the officers monitoring a situation or a risk.

Periodic review of the feedback process, and how the responses are vetted, and actioned will also be done in conjunction with the DMC, and its regional operators to improve and further uphold the ongoing SLAs.

Project Technology Assessment

Currently, the DMC under CResMPA aegis, is conducting a project to establish the current technological capabilities and operational procedures of the DMC with the aim of improving and refurbishing both the ICT and Civil infrastructures of the DMC and the 24 District DMC's. In collaboration with this project, the initial inception and conceptual study of the

DSS will establish the changes that need to be done at the agencies to meet the DSS target technology architecture.

DSS systems capability, the types of decisions it will automate, and its operational capabilities DMC will have to be mapped as part of the technology assessment to ensure the appropriate data input types, timelines, and administrative functions are understood.

The system must incorporate advanced technologies, Artificial intelligence (AI) based on Data Science Techniques and Large Language Models (LLM). The system will be designed with a modular open-source architecture such that it can be easily extended in the future. The system will also include robust ERP and GIS backend systems to ensure scalability, efficiency, and compliance with international best practices.

The Large Language Model (LLM) or a Machine Language-based Rules Engine will be configured with predefined rules to action when risks or issues are detected. The definition of risk and issues based on each type of input from agencies needs to be mapped and tested to ensure appropriate actions, notifications and workflow options have been adopted by the **DSS**. The rules engine also needs to operate in a robust and agile manner when errors, and/or unforeseen risks are encountered.

DSS technology assessment needs to ensure the outputs of the DSS system can be appropriately delivered to all intended parties. This involves an assessment of the recipient's technological capability, the technological knowledge of the operational personnel, and a verification methodology to ensure critical information has been correctly received and understood.

The DSS Ecosystem needs to be assessed to ensure all stakeholder groups are fully onboard and contribute to the functional and Operational Continuity Plan (OCP) of the target operating model of the DSS.

Disaster Recovery (DR), and OCP must be done during this inception phase to identify the critical functions that need to be established to ensure operations can continue at all DMC centers, and at the agencies where the DSS receive its critical information.

All tasks described in this TOR, will be carried out in close consultation with DG DMC and staff and, the Project Coordination Team (PCT).

Stakeholders will have clearly defined responsibilities to ensure effective inter-agency coordination. The Disaster Management Center (DMC) will act as the primary coordinating body, overseeing the development, implementation, and operation of the DSS. The Department of Meteorology (DoM), Irrigation Department (ID), and National Building Research Organization (NBRO) will provide real-time and historical data inputs critical for predictive analytics and situational awareness. Regional and district-level disaster management authorities will be responsible for implementing DSS-driven recommendations and ensuring community-level preparedness. The Ministry of Defence will facilitate high-level coordination and integration across national security and emergency response agencies. All stakeholders will participate in periodic review meetings to ensure alignment with project goals, address implementation challenges, and adapt strategies based on emerging risks and opportunities. Clear communication protocols will be established to enhance collaboration and streamline workflows across agencies.

Scope of Work and Tasks

The total duration of the contract would be 64 weeks.

The work of the consultant shall consider the following areas:

- A. Review existing systems and resources of DMC to identify the gaps and weaknesses to update the existing schematic (concept) and requirements in line with the TOR.
- B. Overview of the existing systems and outputs of other relevant stakeholders to ensure seamless communication from upstream to downstream.
- C. Provide total solutions for the DSS including the following not limited to.
 - i. The consultant shall prepare a comprehensive DSS architectural and engineering plan that facilitates decision-making by emergency and disaster management stakeholders and communities. It enables enriched scenario planning and supports the strategic, tactical, and operational decisions that arise in emergency and disaster events. The platform facilitates the real-time collection of Risk-Vulnerability-Exposure-data, Management, analysis, distribution, and visualization of data and information for enhanced situational awareness at the different administrative levels. Ensure centralized repositories unify multi-source data management. The platform should have data layers preloaded with the solution relevant to disaster management.
 - ii. Data interoperability between agencies must be facilitated through DMC data platform, Ensuring compatibility of data usage across different formats.
 - iii. Integration of location-based data for mapping hazards and vulnerable areas and Geospatial Information Systems (GIS) should be facilitated.
 - iv. Consultant shall provide Impact-based Early Warning generation, Risk visualization, Last-mile Early Warning dissemination system with common alerting protocol (CAP), Public communication through the call center, and Emergency response coordination, to enable comprehensive understanding and optimized solutions for the DSS platform. The solution should feature predictive modelling for natural phenomena, risk assessment tools to estimate impacts on populations and infrastructure and early warning systems with location-based public alerts and real-time updates
 - v. The solution should Include decision-making tools with scenario analysis for evaluating disaster response outcomes, resource allocation modules for critical resources, and visualization tools for rapid situational assessment.
 - vi. The system should provide communication and collaboration tools for synchronizing efforts, along with community engagement tools for disaster management.
 - vii. Consultant shall assess the ICT facilities and communication facilities of DMC (Software, Hardware) and recommend a total solution to achieve an effective, reliable, sustainable system for DSS with an Early Warning dissemination mechanism. Expected dispatch speed: minimum of 0.5 million per minute.
 - viii. Include operational support modules for real-time disaster response management, and unified oversight, and management tools for tracking and coordinated transportation.
 - ix. Stakeholder management and emergency communication, Disaster relief information, reporting local incidents, providing updates on requests, Information on evacuations, generation of situational reports, and continuous communication from the public to support the DSS systems.
 - x. As part of Relief and Recovery Planning, Systems to manage long-term recovery efforts and reconstruction by Data gathering and reporting should be included.
 - xi. The system should facilitate real-time updates and access by field operatives and citizens, featuring geotagged data updates, tracking, notifications, offline data

- collection, route optimization for rescue and relief. Role-based dashboards with customizable interfaces should cater to diverse users.
- xii. Consultant shall prepare a coordination plan with officials in DMC, District authorities, Divisional authorities of other related authorities, and Village level officials and tri-forces, police.
 - xiii. Maintain knowledge repositories and Include tools for disaster preparedness.
 - xiv. The system should enable public information dispatch and support data from crowd feedback to identify disasters. It should also allow the public to upload data /information and provide feedback.
 - xv. The system should enable disaster warnings via TV broadcasters and through telecom service providers, with mechanisms to track news delivery and confirm receipt by GNs, first responders, and the public. Means to assess the effectiveness of alerts and mitigation plans should be provided.
 - xvi. The system must address all disasters outlined in the DMC framework, connect with all relevant agencies, gather data, and facilitate communication and integration with existing and new agency data sources.
 - xvii. Consultant shall develop a user manual to share with the operational-level mechanisms.
 - xviii. The consultant shall complete a pilot run of the DSS project with the participation of authorities' operational-level officials of the DMC, and the responsible officers from each technical agency (Meteorology Department, Irrigation Department, National Building Research Organization (NBRO), and other relevant technical agencies.
 - xix. After concluding the pilot studies, a review will be done to identify the possible improvements to the platform. This review will consist of a community survey to track their opinion on the proposed solution and get their feedback for further improvements.
 - xx. The consultant shall actively work with DMC staff to develop a DSS platform by monitoring all the above objectives and a results-based framework for the DMC component of the CResMPA project.
 - xxi. The proposed application should be preferably a locally developed/customised solution, a locally hosted solution and should consist of a centralized spatial database server with preloaded data layers on a reliable and secure map.
 - xxii. The consultant shall demonstrate the conceptual framework to be aware of the road map, government arrangements, data flow, early warning and feedback mechanism, and other deliverables and tasks.
 - xxiii. It should comprise but not be limited to facilities such as. Simultaneous multi-user access through any internet browser, Access levels, Dashboard customization for levels, Centralized device management, and Web Services extended by browser-side APIs. The system should support, interaction with users, support Geoprocessing and Geolocation IP Addresses, Publish raster, vector and tabular data, and reliable and secure map for API management.
 - xxiv. The platform should include built-in tools for manipulating GIS and base maps, using reliable and secure maps as the base map. It must provide preloaded data layers.

Features should include cartographic inquiry, asset, staff and citizen management and spatial and non-spatial layer management and editing of geometric data and attributes.

- xxv. The Platform should be capable of GIS base analytical functions, Location wise behavior, Proximity, Statistical analysis, Weather Analysis, Tracking Analysis, Planning the route for recoveries, and Data Analysis through customized dashboards
 - xxvi. The Platform should be capable of exporting the map information from Excel, XML, PDF, GeoJSON and KML formats allowing users to print the map in Jpeg formats and be capable of generating different types of reports.
 - xxvii. Basic GIS functionalities such as combining spatial and data queries should be supported.
 - xxviii. The system should be capable of multiple users' simultaneous access, with the ability to upload photos in real-time, display WebGIS application in real-time, auto display data location with available items.
 - xxix. An agreement on the server maintenance and other costs incurred overall of the platform should be outlined by the consultant for a minimum period of eight years after the full implementation of the project.
 - xxx. The consultant shall prepare the required specifications for goods, works, and services for DSS operations based on the World Bank guidelines.
 - xxxi. The consultant shall install, furnish, commission and operationalize, the software components and hardware equipment for smooth operation.
 - xxxii. The consultant shall hand over all software licenses and authentication to the DMC without any other payment and hidden agenda
 - xxxiii. High Availability and Disaster Recovery: The system must ensure high availability (99.999% uptime or greater) through failover clustering and load balancing. Additionally, it must include a comprehensive Disaster Recovery (DR) plan with defined Recovery Time Objectives (RTO) and Recovery Point Objectives (RPO) to maintain operational continuity during system outages or disasters.
 - xxxiv. Proven expertise in AI/ML implementation, ERP system integration, and GIS-based visualization.
 - xxxv. Data Integration and Real-Time Analytics: The system must support robust data integration through standardized to seamlessly ingest real-time and historical data from diverse sources. The solution should include AI/ML-powered analytics modules to provide predictive insights and actionable intelligence for effective decision-making.
 - xxxvi. The vendor should demonstrate their capabilities through implementations and data visualization achieved in a similar project.
 - xxxvii. Configurable Rules Engine with Automated Workflows: The solution should feature a configurable rules engine that allows the creation of dynamic thresholds and workflows. The system must include an automated escalation mechanism that triggers real-time notifications or corrective actions based on user-defined conditions.
- D. Consultant shall provide all necessary infrastructure facilities including Servers, network facilities, storage and any other hardware requirement upon this DSS project for eight years.
- E. Consultant shall provide all necessary training and hands-on training on DSS.

- F. The consultant shall provide 8 years warranty for DSS software including updated versions and 8 years warranty for hardware and all accessories.

Performance Standards

To ensure the success of the Decision Support System (DSS) and its alignment with the project objectives, a set of performance standards will be implemented. These standards will incorporate Key Performance Indicators (KPIs) to measure system efficiency, accuracy, and stakeholder satisfaction.

System Efficiency

- A. **Uptime:** The DSS should achieve at least 99.9% system availability, minimizing downtime during critical periods.
- B. **Response Time:** Data processing and system response times for real-time analytics must not exceed 2 seconds for most queries.
- C. **Scalability:** The system should handle increasing data loads without performance degradation, demonstrated by stress testing at 150% of expected peak usage. (At peak load on the system, system utilization should not cross 66% level)

Accuracy of Predictive Analytics

- D. **Forecast Precision:** Predictive models for floods, landslides, and other hazards must achieve at least 85% accuracy in identifying potential risks.
- E. **False Alarm Rate:** Maintain a false alarm rate of less than 5%, ensuring reliable alerts that do not overwhelm response teams.
- F. **Validation Metrics:** Use statistical measures such as precision, recall, F1 score, and RMSE (Root Mean Square Error) to validate predictive models against historical data.

Stakeholder Satisfaction

- G. **Training Effectiveness:** At least 90% of trained personnel should demonstrate proficiency in using the DSS, assessed through post-training evaluations.
- H. **User Feedback:** Achieve a stakeholder satisfaction rate of at least 85%, based on surveys assessing ease of use, system reliability, and overall effectiveness.
- I. **Community Engagement:** Positive feedback from community representatives during pilot testing and periodic reviews, with specific emphasis on the usability of early warning systems and disaster preparedness tools.

Monitoring and Review

Performance metrics will be reviewed monthly during progress reporting and after major system updates or disaster response events. These KPIs will guide continuous improvement, ensuring that the DSS evolves to meet operational needs and global disaster resilience standards.

Deliverables/ Tasks

Deliverable 1/Task 1: Inception report including detailed project schedule (6 weeks)

The consultant would conduct a full-scale Service Requirement gathering to finalize the need assessment for the development of the DSS for each stakeholder group. Requirement gathering would cover current systems, bottlenecks and wish lists to evaluate during the review stage.

The consultant will provide an overview of the entire assignment including a work plan and detailed project schedule. This task is to be completed within 6 weeks from the commencement of services.

Inception Report:

The consultant's work plan and team organization, assignment of functions for Consultant's personnel, including any changes or additions, agreed upon during negotiations on the signing of the Contract. This report must also clearly specify all risks and issues, which may negatively affect Project deadlines and the effective execution of all works.

Risk Matrix and Mitigation Strategies

As part of the Inception Report, a comprehensive risk matrix will be developed to identify potential challenges, assess their impact and likelihood, and outline mitigation strategies. The risk matrix will focus on ensuring project success by proactively addressing possible obstacles. **A sample matrix is provided below for reference purposes.**

Risk	Likelihood	Impact	Mitigation Strategy
Data Security Breaches	High	High	Implement robust cybersecurity measures, including AES-256 encryption and firewalls. Conduct regular vulnerability assessments and penetration testing.
Delays in Deliverables	Medium	High	Establish clear timelines, monitor progress through monthly reports, and conduct frequent status meetings with stakeholders.
Inconsistent Data Quality	High	Medium	Develop data validation protocols, provide training to data providers, and ensure continuous feedback loops for data accuracy.
Stakeholder Misalignment	Medium	High	Conduct regular coordination meetings, define roles and responsibilities clearly, and implement an escalation process for resolving conflicts.
System Scalability Issues	Low	High	Test the system under peak load conditions during development and ensure a modular, scalable architecture.
Lack of User Adoption	Medium	Medium	Conduct extensive user training, create user-friendly interfaces, and gather feedback during pilot runs to address usability issues.
Technological Challenges	Low	Medium	Perform a technology assessment during the inception phase to identify gaps and plan for upgrades where needed.
Natural Disruptions	Medium	High	Develop a Disaster Recovery (DR) plan, including backup systems and continuity protocols for critical operations.

The risk matrix will be reviewed and updated periodically throughout the project lifecycle.

Deliverable 2/Task 2: Conceptual framework including technology architecture, and Schematic sketches with governance (6 weeks)

Review and update Schematic and Finalize Design and Development

Review of Schematic (Concept): The consultant shall review the schematic systems of DSS. After identifying the required details of the DSS, the consultant shall prepare an action plan for the entire DSS including sketches and diagrams, to develop the schematic and reach an agreement with DMC.

Legal assignments for data/information sharing:

Preparation of MOUs, SLAs and stakeholder agreements.

All relevant MOUs, SLAs, stakeholder agreements, and current best practices and SOPs should be coordinated and signed by the DMC. Those relevant papers and documents should be prepared by the consultant. (this activity should be in line with the DM Act, NDMP, NEOP, DM policy, and Institutional DM plans)

Deliverable 3/Task 3 and 3.1: Design Development and Build up the DSS software with ICT support and Design Development and Build up the mobile API. (30 Weeks)

Based on the Task 2 deliverable, the consultant shall deliver detailed technology architecture, engineering design drawings and high-level architectural designs for DSS and develop the software product.

Deliverable 4/Task 4: Review the communications network with early warning systems for integration with DSS: (4 weeks)

The consultant shall review the current communications network and prepare a plan to integrate the Early Warning dissemination (CAP) system into DSS.

Deliverable 5/Task 5: Data/ Information Management: (4 weeks)

Data collection, entry, validation, and data banking:

The consultant shall undertake all activities related to the data management and design schematic diagrams, and specifications, for related services. Data protection by data breaching. The DMC will coordinate this task based on the requirement.

Deliverable 6/Task 6: Operationalized the DSS and Testing: (4 weeks)

The consultant shall assist DMC in operationalizing DSS. Finally commissioning of the respective solutions and 5 years (Five) maintenance and ensure long-term sustainability of the API DSS solution.

Project Reporting (compulsory)

Monthly and Periodic Progress Reports:

Progress on mitigation strategies will be tracked in monthly reports to ensure risks are effectively managed and do not hinder project outcomes. This proactive approach will help maintain project alignment with World Bank requirements and ensure smooth implementation. The consultant shall prepare of monthly progress report.

Monthly Reports:

The monthly reports should include:

- Description of Project progress in general, problems, and measures to solve them
- Detailed description of Consultant's executed work
- Recommendations on the organization of work for the next reporting period

Analytical Reports

Technical specifications and other relevant documents.

Final Report:

The final report on the performance results of the project should be prepared based on previous reports, with an analysis of execution and interrelation of individual components. This report should include:

- A. General description of all executed work.
- B. Detailed description of the consultant's work performed during the project implementation.
- C. Through performance testing reports, user feedback, and iterative improvements, the consultant shall analysis of performed upgrades and recommendations for future activities to ensure ongoing operational effectiveness of the DSS systems.

All reports, draft reports, and documentation, related to the execution of bidding procedures under World Bank requirements (technical requirements, technical specifications, bidding documentation packages, evaluation, and other reports - as stated above), should be prepared in English.

The final report will be prepared by the Consultant in the English language in hard copies (the number of copies decided by the PCT based on the number of technical review committee members) and in electronic format.

Deliverable 7 /Task 7: Completion Reports including Operational Manuals, source codes and as build drawing:

The consultant shall prepare of final project completion report with operational manuals.

Deliverables

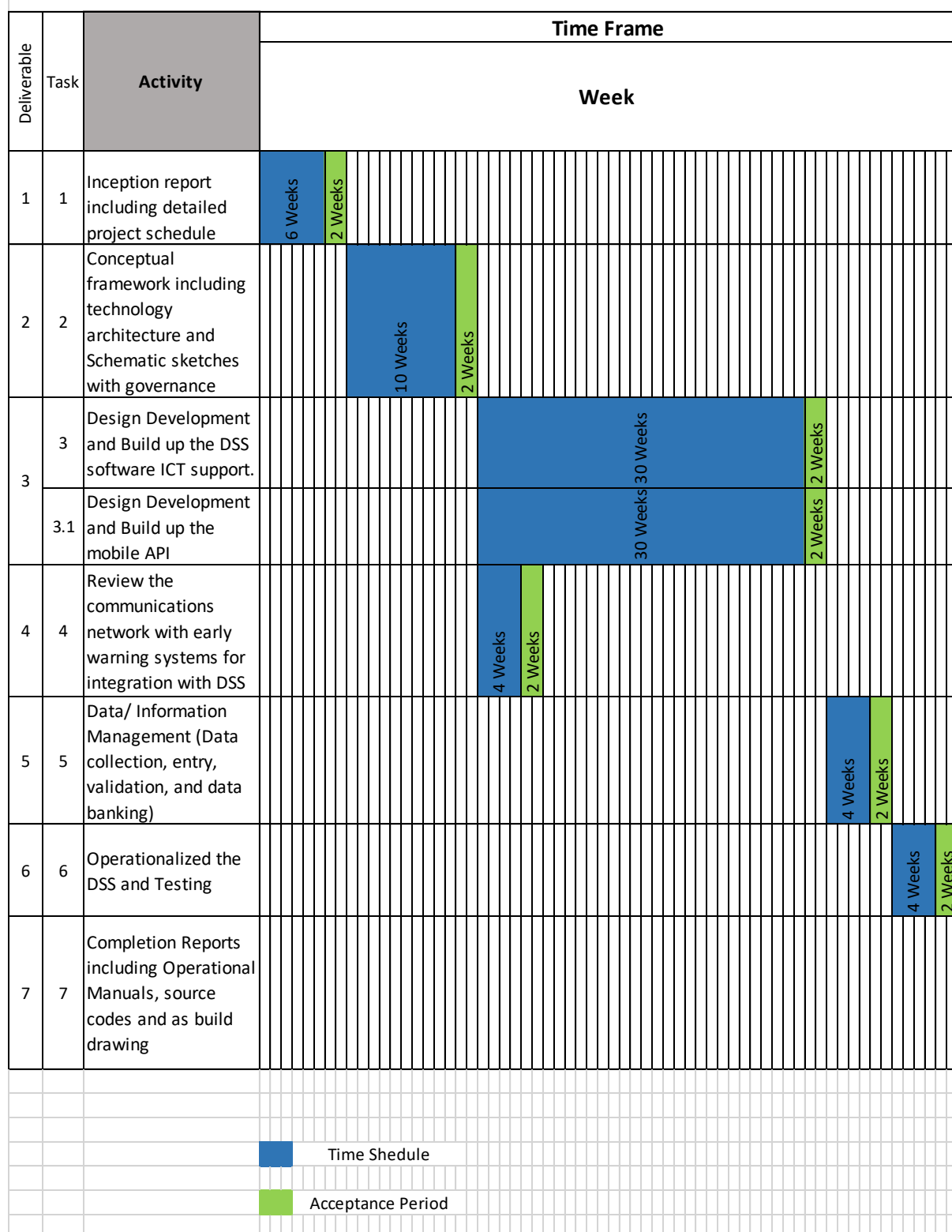
All deliverables shall be prepared in the English language and provided in hard copies and electronic format. All data and analysis developed in the context of this assignment will be provided to the government in an open electronic format (graphics and design reports, etc.) and will be the property of the government. Tasks should be completed within 64 weeks months after contract signing.

Deliverable	Task	Output	Time Schedule
1	1	Inception report including detailed project schedule	6 weeks after the commencement of services. (Accepted within 2 weeks)
2	2	Conceptual framework including technology architecture and Schematic sketches with governance	10 weeks from the acceptance of the Inception Report. (Accepted within 2 weeks)
3	3	Design Development and Build up the DSS software ICT support.	30 weeks from acceptance of deliverable 2. (Accepted within 2 weeks)
	3.1	Design Development and Build up the mobile API	
4	4	Review the communications network with early warning systems for integration with DSS	4 weeks from acceptance of deliverable 2. (Accepted within 2 weeks)
5	5	Data/ Information Management (Data collection, entry, validation, and data banking)	4 weeks from acceptance of Task 3. (Accepted within 2 weeks)
6	6	Operationalized the DSS and Testing	4 weeks from completion of Task 5. (Accepted within 2 weeks)
7	7	Completion Reports including Operational Manuals, source codes and as build drawing	The completion report shall be submitted 2 Weeks after testing and operationalization of the software and the Operational Manual.

Note- Endorsement of the reporting document and deliverable is performed by the Project Director within 2 weeks from the submission of documents to the PCT. The consultant shall perform the task and deliverables on a rolling schedule.

(Provide a clear Gantt chart for visual clarity. And include a detailed financial section specifying payment linked to milestones. This can be specified in percentages of the total bid amount.)

Gantt Chart for Deliverables and Tasks



Payment Schedule

Mobilization Advance of 10% of the total Contract amount in the relevant currencies is paid upon submission of the advance payment guarantee and amortization shall commence from the initial payment and be completed as financial progress has reached 50%.

No	Deliverables	Percentage
1	Inception report including detailed project schedule	5%
2	Conceptual framework including technology architecture and Schematic sketches with governance	10%
3	Design Development and Build up the DSS software ICT support.	30%
4	Review the communications network with early warning systems for integration with DSS	10%
5	Data/ Information Management (Data collection, entry, validation, and data banking)	15%
6	Operationalized the DSS and Testing	10%
7	Completion Reports including Operational Manuals, source codes and as build drawing	20%

General Requirements of the Consultant

To align with global standards set by organizations and the World Bank.

the Consultant

- A. The consultant should have fair knowledge in the assessment and analysis of natural disasters within Sri Lanka - experience working with the government of Sri Lanka and DRR activities would be highly beneficial.
- B. The consultant should work full-time to achieve the objectives within the consultancy period.
- C. The consultant should have close coordination with the ongoing CResMPA (SLMO/CRES/DMC/C/QCBS/41-1) project in the DMC to facilitate the infrastructure and further planning activities and also to avoid any miscommunications.

Software Development Life Cycle

The software development life cycle (SDLC), is a structured process used to design, develop, test, and deploy software applications. To achieve the objectives of this project, the consultant should have the following staff with experience.

Consultant Selection Criteria

To align with SL guidelines and the World Bank the following criteria will be used to evaluate consultant proposals:

Technical Experience and Expertise

Personnel

1. Planning and Requirement Analysis

Objective: Determine the project's goals and gather requirements.

Activities: Feasibility study, project planning, and defining the software requirements.

Professionals Involved:

Project Manager/Team Leader:

Responsibilities: Project managers monitor the implementation of defined processes and frameworks. The project manager ensures that they follow its methodology. Aside from the SDLC, they ensure that processes in terms of communication and assignments are well-established. Data and information gathering, analysis, and analysis of detailed requirements. Preparation of specifications, user stories, use cases, and process flows. Ensure the requirements align with business goals and are clearly understood by the development team. Overall coordination, planning and overseeing the project timeline and resources to achieve project objectives.

Qualifications: With at least 10 years experience managing IT-related projects. Must have proven experience in agile software development methods and understanding of new Development and Operational processes with relevant project management qualifications. PMP certification for project management.

Involvement: Acts as a liaison between business stakeholders and the technical team, ensuring that all parties have a mutual understanding of the project scope

Business Analyst:

Responsibilities: The BA's job is to take in all requests, figure out why people have made them, evaluate and rank the results of that analysis, and then assign priorities to the requests. A BA's ability to think critically is a key for determining which features are most important to the business and what can be dropped.

Qualifications: With at least 8 years of experience in business analysis. A business analyst has to understand clients' needs and analyze the company's specifics and processes. It's the way to make sure that the project delivery is to be achieved. Understanding of new Development and Operational processes with relevant project management qualifications. IIBA Certification in software product development.

Involvement: Acts as a liaison between business stakeholders and the technical team, ensuring that all parties have a mutual understanding of the project scope.

AI/ML Specialist:

Responsibilities: Based on required specifications, lead program teams to develop the machine learning / Large Language models to be used in predictive analytics for the DSS

Qualifications: 5+ years' experience in developing predictive analytics and machine learning models. A postgraduate degree such as an MSc in AI, Data Science, ML or Analytics is preferred from a globally recognized university.

Subject Specialist:

Responsibilities: Providing domain-specific expertise to ensure accurate modeling of disaster scenarios and validating the relevance and reliability of system outputs. They collaborate with technical teams to align system functionality with real-world disaster management needs.

Qualifications: 05+ years of experience as an executive managing in the Climate Resiliency or Disaster Management domains.

GIS Specialist:

Responsibilities: Integrating geospatial data into the decision support system, ensuring accurate mapping and spatial analysis for disaster management. They develop and maintain GIS layers, tools, and visualizations to support decision-making during disaster scenarios.

Qualifications: 5+ years of Proficiency in GIS/QGIS and geospatial analysis.

Front End Specialist:

Responsibilities : Designing and developing user-friendly interfaces for the decision support system, ensuring seamless interaction and accessibility. They focus on creating intuitive dashboards and visual elements to effectively present disaster management data.

Qualifications: 5+ years of Proficiency in developing robust and secure front-end web applications using industry-standard open-source content management systems (CMS).

Documentation Writer:

Responsibilities : Creating clear, comprehensive, and user-friendly documentation for the decision support system, including user guides, technical manuals, and training materials. They ensure all stakeholders can effectively understand and utilize the system's features and workflows.

Qualifications: 5+ years of proficiency in creating API documentation and user manuals

Vulnerability Assessment & Penetration Testing (VAPT):

Responsibilities: Identifying and addressing security vulnerabilities in the decision support system to ensure robust protection against cyber threats. They conduct regular penetration tests and provide recommendations to enhance system resilience for disaster management operations.

Qualifications: 5+ years of proven proficiency in Vulnerability Assessment & Penetration Testing.

A minimum of B.Sc. in IT, ICT or other relevant subject matter from a recognized Sri Lankan or accredited foreign university (with the University Grants Commission) is necessary for all the key personnel.

2. Design

Objective: Create the architecture and detailed design of the software.

Activities: System and software design, including database design, user interface design, and API design.

Professionals Involved:

Software Architect:

Responsibilities: Provides oversight during development, ensuring that the implementation aligns with the architectural design and standards.

Qualifications: 15+ years of proven proficiency in software architecture and design. BSc in Computer Science or Higher qualifications.

Involvement: Reviews code and guides developers to ensure that the architecture is correctly implemented.

Software Engineers :

Responsibilities: Implements the software system on the design specifications, ensuring responsiveness and user interactivity.

Qualifications: 8+ years of proven proficiency in Experience in Programming Languages, Frameworks and Libraries, Software Development Lifecycle (SDLC), API Development. Testing and Debugging. BSc in Computer Science or Higher qualifications.

Involvement: Writes code for the application, with integrated development environments (IDEs) and development of the systems and APIs according to design.

Front-End Developers:

Responsibilities: Implements the user interface based on the design specifications, ensuring responsiveness and user interactivity.

Qualifications: 6+ years of proven proficiency in Experience with modern front-end frameworks and UI/UX principles. BSc in Computer Science or Higher qualifications.

Involvement: Writes code for the client-side of the application, integrating with back-end services, as necessary.

Back-End Developers/ Mobile API developer:

Responsibilities: Develops server-side logic, creates, and manages databases, and ensures seamless integration between the front-end and back-end.

Qualifications: 8+ years of proven proficiency in Experience with Frameworks, DBs , APIs, Security, testing and debugging. BSc in Computer Science or Higher qualifications.

Involvement: Works on the business logic, Mobile APIs, and data management, ensuring that the application functions as intended.

DevOps Engineers:

Responsibilities: Sets up and manages development environments, automates build processes, and integrates continuous integration/continuous deployment (CI/CD) pipelines.

Qualifications: 6+ years of proven proficiency in Experience with Programming and Scripting, CI/CD Pipelines, Containerization and Orchestration, Security, and Automation. BSc in Computer Science or Higher qualifications.

Involvement: Ensures that the development process is smooth, and that code is regularly tested and deployed in a consistent manner.

Data Engineers

Responsibilities: Designs data architecture, including data pipelines, storage solutions, and integration with various data sources. Ensures that data is collected, processed, and stored efficiently and securely.

Qualifications: 10+ years of proven proficiency in Experience with Data Warehousing and ETL Tool, Databases, Data Pipelines and Workflow Automation, Data Modeling and Architecture, BSc in Data Engineering or equivalent or higher qualifications.

Involvement: Works with the software architect and developers to design data models and pipelines that align with the system architecture. Provides input on data-related requirements and challenges.

Geospatial Consultant

Responsibilities: Designs geospatial components of the software, such as mapping features, spatial data handling, and integration with geographic information systems (GIS).

Qualifications: 15+ years of proven proficiency in Experience with GIS Software Expertise, Spatial Data Analysis, Programming and Scripting, Web Mapping and Domain Knowledge. BSc in Computer Science and Masters qualification or higher qualifications.

Involvement: Collaborates with the software architect and UI/UX designer to integrate geospatial data and functionality into the system design. Provides technical guidance on handling geospatial data and optimizing spatial analysis.

3. Implementation (Coding)

Objective: Convert the design into a functional software application.

Activities: Writing code, developing the user interface, and implementing the backend.

Professionals Involved:

Software Architect

Responsibilities: Provide oversight during development, ensuring that the implementation aligns with the architectural design and standards.

Involvement: Reviews code and guides developers to ensure that the architecture is correctly implemented.

Software Engineers :

Responsibilities: Implements the software system on the design specifications, ensuring responsiveness and user interactivity.

Involvement: Writes code for the application, with integrated development environments (IDEs) and development of the systems and APIs according to design.

Front-End Developers (Varying Experience/Expertise – Senior/Junior Engineers)

Responsibilities: Implements the user interface based on the design specifications, ensuring responsiveness and user interactivity.

Involvement: Writes code for the client side of the application, integrating with back-end services and ensuring a seamless user experience.

Back-End Developers (Varying Experience/Expertise – Senior/Junior Engineers)

Responsibilities: Develops server-side logic, creates, and manages databases, and ensures seamless integration between the front-end and back-end.

Involvement: Works on the business logic, APIs, and data management, ensuring that the application functions as intended.

Data Engineers

Responsibilities: Develops data pipelines, storage solutions, and data processing frameworks to handle large volumes of data. Ensures data integrity, security, and performance.

Involvement: Works closely with back-end developers to integrate data pipelines with the overall system. Develops ETL (Extract, Transform, Load) processes and optimizes database performance.

DevOps Engineers

Responsibilities: Sets up and manages development environments, automates build processes, and integrates continuous integration/continuous deployment (CI/CD) pipelines.

Involvement: Ensures that the development process is smooth, and that code is regularly tested and deployed in a consistent manner.

Project Manager/ Team Leader

Responsibilities: Reviews the test results to ensure that the product meets the acceptance criteria and is ready for deployment.

Involvement: Collaborates with QA and development teams to ensure that all critical issues are resolved before deployment.

Geospatial Engineer

Responsibilities: Implements geospatial features, including mapping functionalities, spatial data processing, and integration with geographic information systems (GIS).

Qualifications: 15+ years of proven proficiency in Experience with GIS Software Expertise, Spatial Data Analysis, Programming and Scripting, Web

Mapping and Domain Knowledge . BSc in Computer Science and Masters qualification or higher qualifications.

Involvement: Collaborates with front-end and back-end developers to integrate geospatial data into the application. Develops custom geospatial algorithms and ensures the accurate display and processing of spatial information.

4. Testing

Objective: Ensure the software is free of defects and meets the requirements.

Activities: Unit testing, integration testing, system testing, and user acceptance testing.

Professionals Involved:

Quality Assurance (QA) Engineers

Responsibilities: Tests the software for defects, performance issues, and overall quality, using both manual and automated testing methods.

Qualifications: 10+ years of proven proficiency in Experience with Software Quality Assurance (QA) with knowledge on Testing Tools, Automation frameworks, bug tracking etc. . BSc in Computer Science or equivalent or higher qualifications. ISTQB CTAL certifications.

Involvement: Executes test cases, reports bugs, and works with developers to resolve issues.

Front-End and Back-End Developers

Responsibilities: Conduct unit testing on the code they have written and collaborate with QA to fix identified bugs.

Qualifications: 8+ years of proven proficiency in Experience with Frameworks, DBs, APIs, Security, testing and debugging. BSc in Computer Science or Higher qualifications.

Involvement: Writes tests for their code, responds to bug reports, and refines the software based on testing feedback.

Data Engineers

Responsibilities: Ensures that data pipelines and storage solutions are thoroughly tested for performance, reliability, and scalability. Validates that the data processing workflows function as expected.

Qualifications: 10+ years of proven proficiency in Experience with Data Warehousing and ETL Tool, Databases, Data Pipelines and Workflow Automation, Data Modeling and Architecture. BSc in Data Engineering or equivalent or higher qualifications.

Involvement: Collaborates with QA to test data integrity, performance, and the effectiveness of Extract, Transport, and Load (ETL) processes. Identifies and resolves data-related issues.

DevOps Engineers

Responsibilities: Manages the testing environment and CI/CD pipelines, ensuring that automated tests are run as part of the deployment process.

Qualifications: 6+ years of proven proficiency in Experience with Programming and Scripting, CI/CD Pipelines, Containerization and Orchestration, Security, and Automation. BSc in Computer Science or Higher qualifications.

Involvement: Facilitates the running of tests and ensures that the environment is properly configured for accurate testing.

Security Specialist

Responsibilities: Conduct security testing to identify vulnerabilities and ensure the software is secure.

Qualifications: 10+ years of proven proficiency in Experience with Security tools, Encryption and Authentication, Compliance Standards. Cloud Security and Penetration testing and Governance. BSc in Computer Science / Cyber Security or higher qualifications. CISSP Certification.

Involvement: Performs penetration testing and reviews the software for potential security issues.

Project Manager/ Team Leader

Responsibilities: Reviews the test results to ensure that the product meets the acceptance criteria and is ready for deployment.

Involvement: Collaborates with QA and development teams to ensure that all critical issues are resolved before deployment.

5. Deployment

Objective: Release the software to users.

Activities: Deployment planning, environment setup, and actual deployment.

Professionals Involved:

DevOps Engineers

Responsibilities: Manages the deployment process, including setting up the production environment, automating the deployment, and ensuring that the software is released smoothly.

Involvement: Executes the deployment, monitors the release, and handles any issues that arise during deployment.

Software Architect

Responsibilities: Oversees the deployment to ensure that the software performs as expected in the live environment and adheres to architectural standards.

Involvement: Provides guidance during deployment and may assist in resolving any technical challenges.

Front-End/Back-End Developers

Responsibilities: Supports the deployment by managing server-side configurations, database migrations, and backend infrastructure.

Involvement: Prepares the backend systems for deployment and ensures they integrate seamlessly with the rest of the environment.

Data Engineers

Responsibilities: Manages the deployment of data pipelines and storage systems, ensuring data flows correctly and that all data components are fully operational in the production environment.

Involvement: Monitors data processing in the live environment, resolving any issues that arise with data handling or ETL processes during deployment.

QA Engineers

Responsibilities: Conducts the final acceptance testing in the production environment to ensure the software is functioning as expected post-deployment.

Involvement: Validates the deployed software and reports any issues that might have occurred during the transition to production.

Project Manager/ Team Leader

Responsibilities: Reviews the deployed product to ensure it meets business requirements and is ready for end-users.

Involvement: Signs off on the release, making the final decision to move the software into production.

6. Maintenance and Operations

Objective: Provide ongoing support, fix issues, and enhance the software.

Activities: Bug fixing, performance tuning, and adding new features.

Projected Man Months (MM) Requirement.

S/N	Position	Qualification requirement
		1. Planning and Requirement Analysis
1	Project/ Business Analyst: (6MM) 2 Nos.	With at least 8 years of experience in business analysis. A business analyst has to understand clients' needs and analyze the company's specifics and processes. It's the way to make sure that the project delivery is to be achieved. Understanding of new Development and Operational processes with relevant project management qualifications. IIBA Certification in software product development.
2	Team leader / Project Manager: (12MM) 2 Nos.	15+ years of proven proficiency in software architecture and design, Design and Architecture Patterns, API Design, Scalability and Performance Optimization. BSc in Computer Science or Higher qualifications.
		2. Design
3	System Architect: (4MM) 1 Nos.	15+ years of proven proficiency in software architecture and design. BSc in Computer Science or Higher qualifications.
4	UI/UX Designer: (8MM) 3 Nos.	6+ years of proven proficiency in Experience with modern front-end frameworks and UI/UX principles. BSc in Computer Science or Higher qualifications.

5	Database Administrator (DBAs): (4MM) 2 Nos.	10+ years of proven proficiency in Experience with Data Warehousing and ETL Tool, Databases, Data Pipelines and Workflow Automation, Data Modeling and Architecture. BSc in Data Engineering or equivalent or higher qualifications.
6	API Designers: (3MM) 1 Nos.	6+ years of proven proficiency in Experience in Programming Languages, Frameworks and Libraries, Software Development Lifecycle (SDLC), API Development. Testing and Debugging. BSc in Computer Science or Higher qualifications.
		3. Implementation (Coding)
7	Software Developer: (16MM) 6 Nos.	6+ years of proven proficiency in Experience in Programming Languages, Frameworks and Libraries, Software Development Lifecycle (SDLC), API Development. Testing and Debugging. BSc in Computer Science or Higher qualifications.
8	Frontend Developer: (12MM) 3 Nos.	6+ years of proven proficiency in Experience with modern front-end frameworks and UI/UX principles. BSc in Computer Science or Higher qualifications.
9	Backend Developers: (12MM) 3 Nos.	8+ years of proven proficiency in Experience with Frameworks, DBs , APIs, Security, testing and debugging. BSc in Computer Science or Higher qualifications.
10	Data Engineers : (8MM) 3 Nos.	10+ years of proven proficiency in Experience with Data Warehousing and ETL Tools, Databases, Data Pipelines and Workflow Automation, Data Modeling and Architecture . BSc in Data Engineering or equivalent or higher qualifications.
		4. Testing
11	Quality Assurance (QA) Engineer: (4MM) 1 No.	10+ years of proven proficiency in Experience with Software Quality Assurance (QA) with knowledge on Testing of Tools, Automation frameworks, bug tracking etc. BSc in Computer Science or equivalent or higher qualifications. ISTQB CTAL certifications.
12	Tester: (6MM) 3 Nos.	3+ years of proven proficiency in operational work on ICT and Software systems testing. BSc in Computer Science or Higher qualifications.
		5. Deployment
13	DevOps Engineer: (8MM) 4 Nos.	6+ years of proven proficiency in Experience with Programming and Scripting, CI/CD Pipelines, Containerization and Orchestration, Security, and Automation. BSc in Computer Science or Higher qualifications
14	System Administrator: (8MM) 3 Nos.	3+ years of proven proficiency in operational work on ICT systems administration. BSc in Computer Science or Higher qualifications.
		6. Maintenance and Operations

15	Service Agreement for eight years.	
		7. API-Specific Considerations:
16	API Designer: (6MM) 2 Nos.	6+ years of proven proficiency in Experience in Programming Languages, Frameworks and Libraries, Software Development Lifecycle (SDLC), API Development. Testing and Debugging. BSc in Computer Science or Higher qualifications.
17	Backend Developer: 3 Nos	8+ years of proven proficiency in Experience with Frameworks, DBs , APIs, Security, testing and debugging. BSc in Computer Science or Higher qualifications.
18	Security Specialist: (6MM) 2 Nos.	10+ years of proven proficiency in Experience with Security tools, Encryption and Authentication, and Compliance Standards. Cloud Security and Penetration Testing and Governance. BSc in Computer Science / Cyber Security or higher qualifications. CISSP Certification.
19	Documentation Writer: (3MM) 1 Nos.	5+ years of proficiency in creating API documentation and user manuals. BSc or equivalent qualification in the related field.
		8. Non-Key Experts:
20	Data Entry Operators (6MM)	1+ years of proficiency and experience in data entry operations and data collection.
21	Data Collectors (6MM)	1+ years of proficiency and experience in data entry operations and data collection
		9. Overhead Cost
22	Training and Awareness (6MM)	Provide all relevant training for new assignments.
23	Development of Operational Manual (3 MM)	Development, and printing of hard copies for relevant stakeholders and DDMCUs
24	Network Facilities	Provide all network diagrams, Hard and soft copies.

Collaboration and Communication

Throughout the SDLC, collaboration and communication are critical. By involving the right professionals at each stage and ensuring effective communication and collaboration, the software development cycle can be executed efficiently, resulting in high-quality API software.

Collaborative Tools and Platforms for Effective Communication in the SDLC

To facilitate seamless collaboration and communication throughout the software development life cycle (SDLC) of the Decision Support System (DSS), a range of advanced tools and platforms will be employed. These tools ensure efficient coordination among stakeholders, real-time updates, and streamlined workflows.

Communication and Collaboration

- A. For real-time messaging, video conferencing, and file sharing, ensuring quick decision-making and effective coordination among agencies and team members.

- B. For virtual meetings, stakeholder consultations, and training sessions, especially for geographically dispersed teams.
- C. Visual collaboration on system architecture, workflow designs, and brainstorming sessions.

Project Management

- D. For agile project management, including sprint planning, task tracking, and issue resolution.
- E. For task allocation, progress tracking, and collaboration on project deliverables.
- F. For detailed project scheduling, resource management, and Gantt chart creation.

Document Sharing and Version Control

- G. For centralized document storage, ensuring version control and collaborative editing of project documentation.
- H. As a knowledge-sharing platform to document decisions, workflows, and project progress.

System Development and Testing

- I. For collaborative source code management, version control, and peer reviews.
- J. For continuous integration and deployment (CI/CD) pipelines to streamline development and testing processes.
- K. For managing test cases, tracking test results, and reporting on system quality.

Feedback and Reporting

- L. For tracking feedback from stakeholders, managing issue logs, and generating status reports.
- M. For creating visual dashboards to present project metrics, progress, and analytics in an accessible format.

Stakeholder Engagement

- N. For real-time surveys, polls, and feedback collection during workshops or consultations.
- O. For structured feedback collection and analysis post-training sessions or project milestones.

Data Sharing and Visualization

- P. For geospatial data visualization and mapping, enabling collaborative disaster risk assessment and resource planning.
- Q. For real-time data streaming and integration across stakeholder systems, ensuring up-to-date situational awareness.

Security and Access Control

- R. For secure identity management and access control, ensuring only authorized personnel can access specific systems or data.

Workflow Automation

- S. For integrating various tools and automating repetitive tasks, such as status updates and email notifications.

Feedback and Continuous Improvement

- T. To gather feedback on tool usage, stakeholder satisfaction, and areas for improvement.

By leveraging these collaborative tools and platforms, the project will ensure efficient communication, minimize delays, and promote transparency throughout the SDLC. This approach will foster alignment among diverse stakeholders, contributing to the successful implementation of the DSS.

Implementation Arrangements

The consultant will be hired on behalf of the DMC by the PCT. The DMC will provide:

- A. Office space including normal office furniture in the DMC premises in Colombo to facilitate close collaboration between DMC staff and the consultant, to prepare documentation, and any other work related to the assignment.
- B. DMC will provide all required liaison (DPD-DMC), documents, correspondence, and any other information relating to the project and deemed necessary for accomplishing the assignment.
- C. No transport facilities will be provided to the consultant, the consultant will arrange their transportation for their work.

Consultant's Responsibilities

The consultant will use accepted financial, administrative (legal), and management practices and will be accountable to the DMC/PCT following the terms of the tender and contract.

For commitment and proper interaction of all individual components of the system to be designed and supervised by the consultant, here defined as the General Consultant for clarity, the following official statement must be issued for all requests for proposals and invitation documents:

"All activities of selected subcontractors under the current tender are subject to monitoring, control, and audit by the DMC/PCT. The General Consultant will be assisting the DMC and CResMPA PCT in the execution of this assignment. Accordingly, the supplier/subcontractor must comply with the monitoring and audit procedures to be determined by the DMC and CResMPA PCT and General Consultant".

The above-listed aspects should ensure necessary actions on the execution of the integration of the system. Subcontractors will be subject to periodic checks of work performed, reports, and procured goods and services for their conformity with system requirements and specifications.

Additional Reporting Requirements

To ensure consistency and clarity in communication, standardized reporting formats will be utilized for all deliverables. These formats will include templates for inception, progress, and final reports, specifying required sections such as objectives, methodologies, findings, challenges, and recommendations. Each report will follow a uniform structure, including an executive summary, detailed narrative sections, visual aids (e.g., graphs, charts, and maps),

and appendices for supplementary information. These templates will be shared with all stakeholders at the outset of the project to streamline documentation processes and ensure alignment with World Bank standards for transparency and accountability. Additionally, all reports will be prepared in both hard and electronic copies, ensuring accessibility and ease of review.

Expected Deliverables and Outputs and Approval of Reporting Documents and Final Reports

- A. Approval of Reporting documents and deliverables is performed according to the following procedure:
- B. Endorsement of Reporting documents and deliverables is performed by the Project Director (PD) within 2 weeks from the handover of Reporting documents to PCT.
- C. Approval of Final Reports is performed by the Project Director within 2 weeks from the handover of Final Reports to PCT.

Monitoring and Evaluation (M&E) Plan

A comprehensive Monitoring and Evaluation (M&E) plan will be developed to track the progress and effectiveness of the Decision Support System (DSS) project. This plan will include specific indicators to measure outputs, outcomes, and impacts, ensuring accountability and alignment with project objectives.